

Unit 6 : Deep Learning in Sentiment Analysis

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Introduction to Sentiment Analysis

What is Sentiment Analysis?

Sentiment Analysis is a technique used in **Natural Language Processing (NLP)** and **Artificial Intelligence (AI)** to identify emotions, opinions, and attitudes from text data.

It helps computers understand:

- Positive emotions 😊
- Negative emotions 😡
- Neutral statements 😐

Sentence	Sentiment
"This phone is amazing!"	Positive
"The service was terrible."	Negative
"The product arrived yesterday."	Neutral

Why Sentiment Analysis is Important?

Huge amounts of text are generated every day through:

Social media

Reviews

Blogs

News

Online shopping websites

Companies use sentiment analysis to:

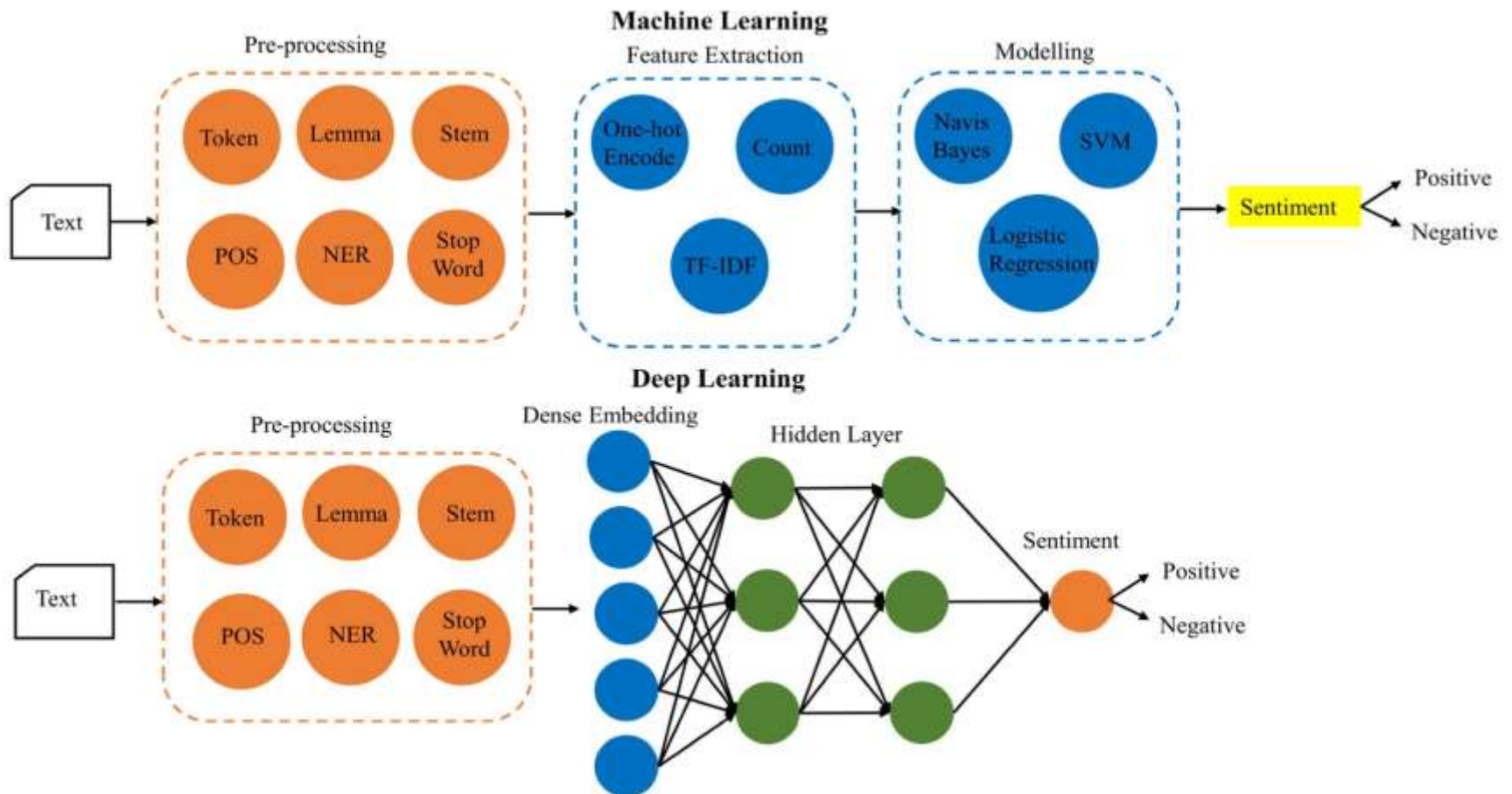
Understand customer feedback

Improve products

Monitor brand reputation

Predict customer behavior

Why Sentiment Analysis is Important?



Need of Deep Learning in Sentiment Analysis

Why Traditional Methods Failed?

Traditional machine learning methods:

- Required manual feature extraction
- Could not understand context properly
- Failed with sarcasm and complex language
- Had lower accuracy

Traditional Algorithms:

- Naive Bayes
- SVM
- Decision Tree

Need of Deep Learning in Sentiment Analysis

Why Deep Learning?

Deep Learning

automatically learns:

- Important features
- Semantic meaning
- Contextual understanding
- Emotional relationships

Advantages:

- Better accuracy
- Handles large datasets
- Learns automatically
- Understands sequence and context

Example

Sentence:

“The movie was unexpectedly good.”

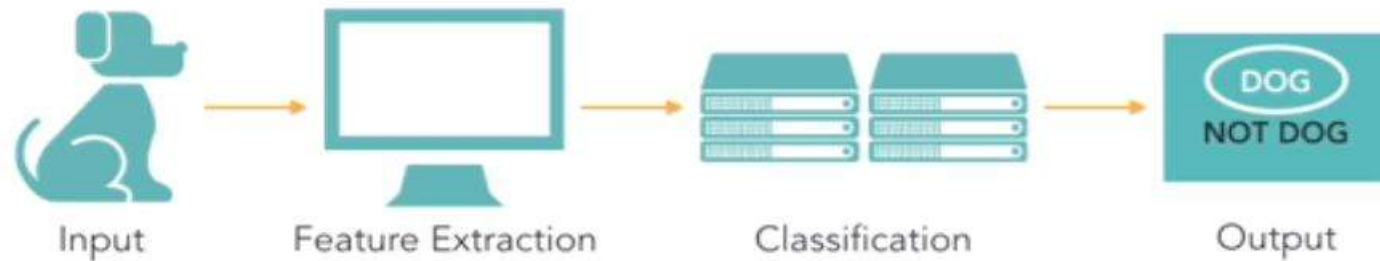
Traditional methods may focus on “unexpectedly”.

Deep Learning understands:

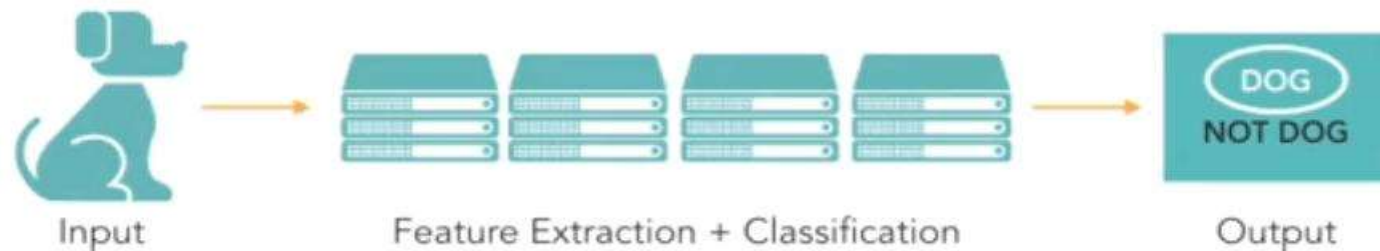
- “good” expresses positivity.

Need of Deep Learning in Sentiment Analysis

TRADITIONAL MACHINE LEARNING



DEEP LEARNING



Deep Learning Architecture for Sentiment Analysis

Basic Workflow

1. Input Text
2. Text Preprocessing
3. Word Embedding
4. Neural Network Processing
5. Sentiment Prediction

Steps in Detail

1. Text Collection

Data collected from:

- Twitter
- Amazon Reviews
- Facebook
- News comments

2. Preprocessing

- Remove punctuation
- Remove stopwords
- Tokenization
- Stemming/Lemmatization

3. Word Embedding

Converts words into vectors.

4. Deep Learning Model

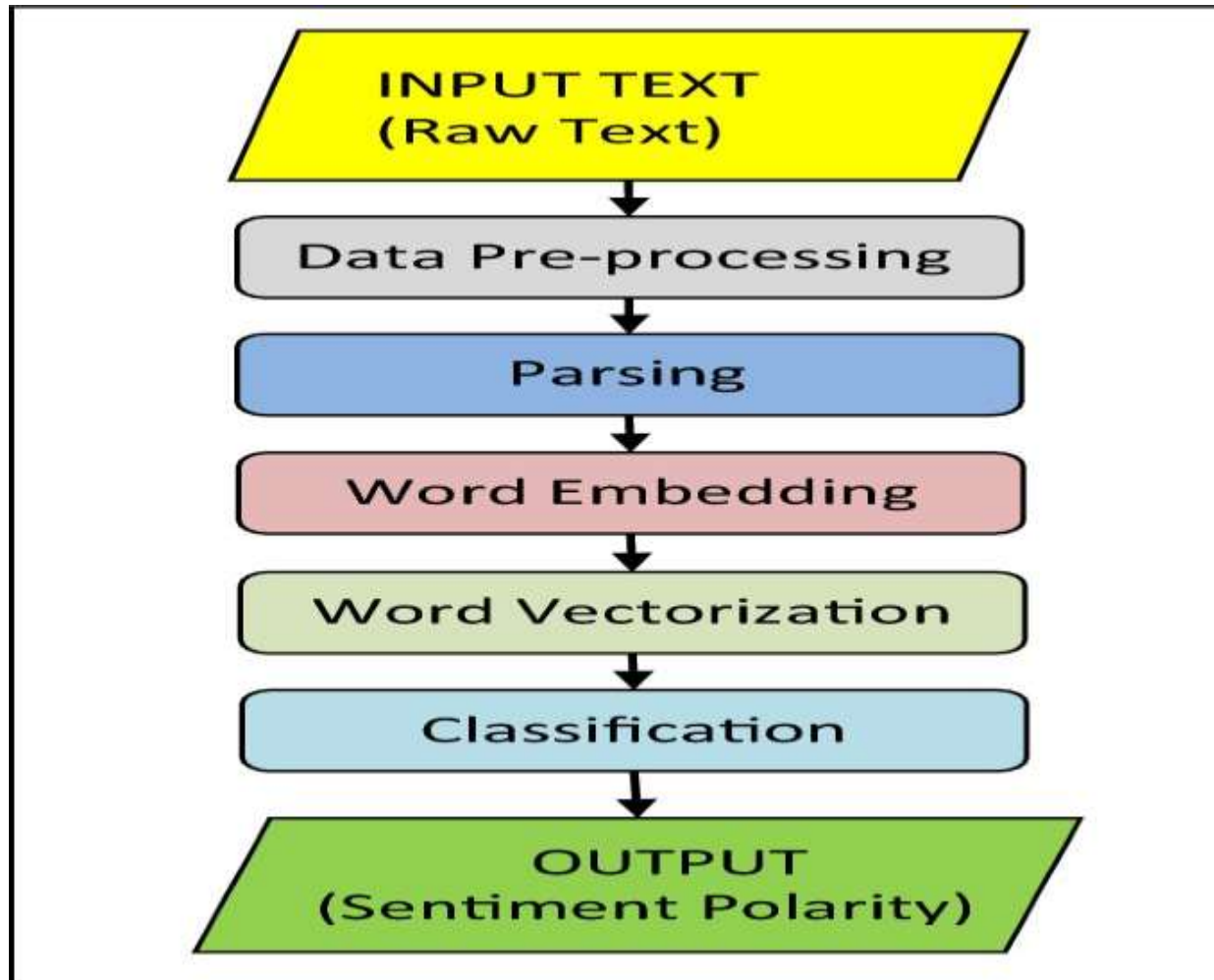
CNN / RNN / LSTM analyzes patterns.

5. Output

Predicts sentiment:

- Positive
- Negative
- Neutral

Deep Learning Architecture for Sentiment Analysis



Sentiment-Specific Word Embedding

What is Word Embedding?

Word Embedding converts words into numerical vectors.

Computers cannot understand text directly.

They understand numbers.

Word	Vector Representation
Happy	[0.21, 0.78, 0.91]
Sad	[0.92, 0.11, 0.33]

Words with similar meanings have similar vectors.

Sentiment-Specific Word Embedding

Popular Word Embedding Methods Sentiment-Specific Embedding

Word2Vec

Learns word relationships.

GloVe

Uses statistical information.

FastText

Understands subwords.

Normal embedding:

- “good” and “bad” may appear close due to context.

Sentiment embedding:

- Separates positive and negative words properly.

Positive Words

Excellent

Amazing

Fantastic

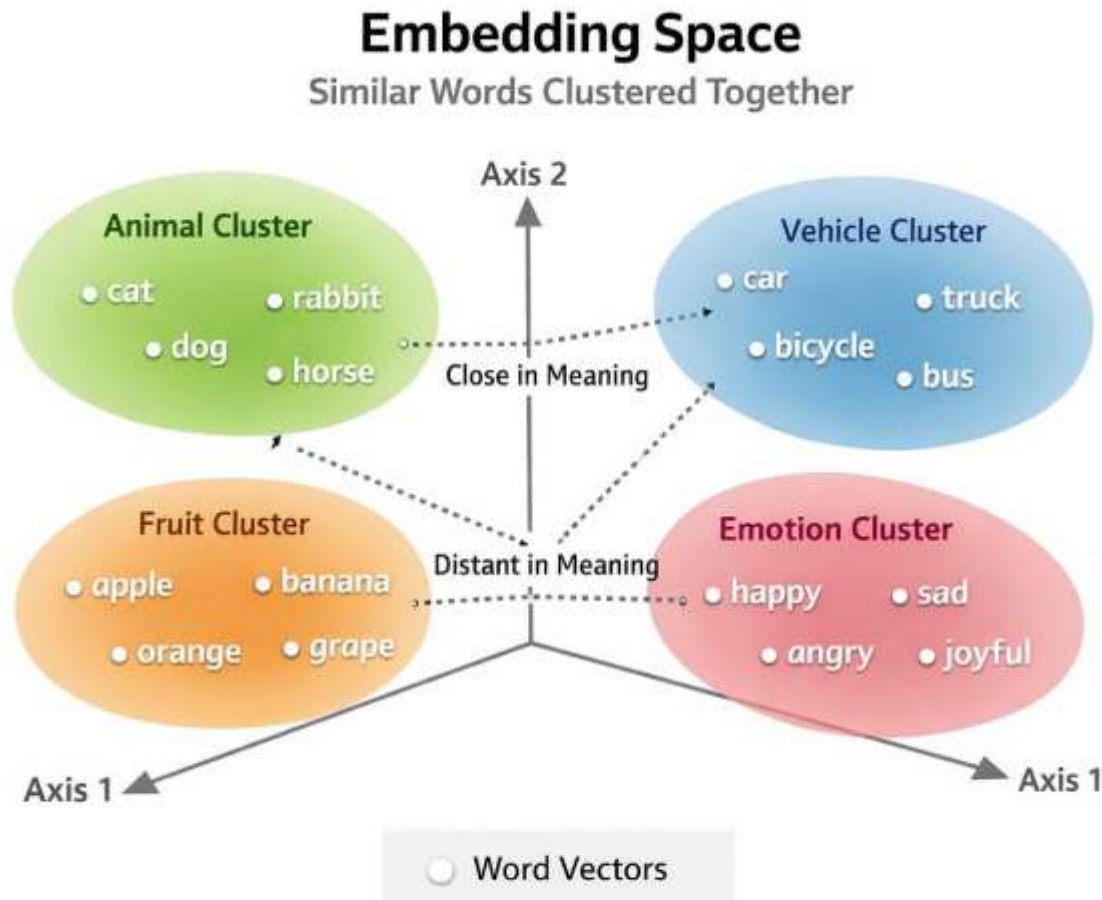
Negative Words

Horrible

Terrible

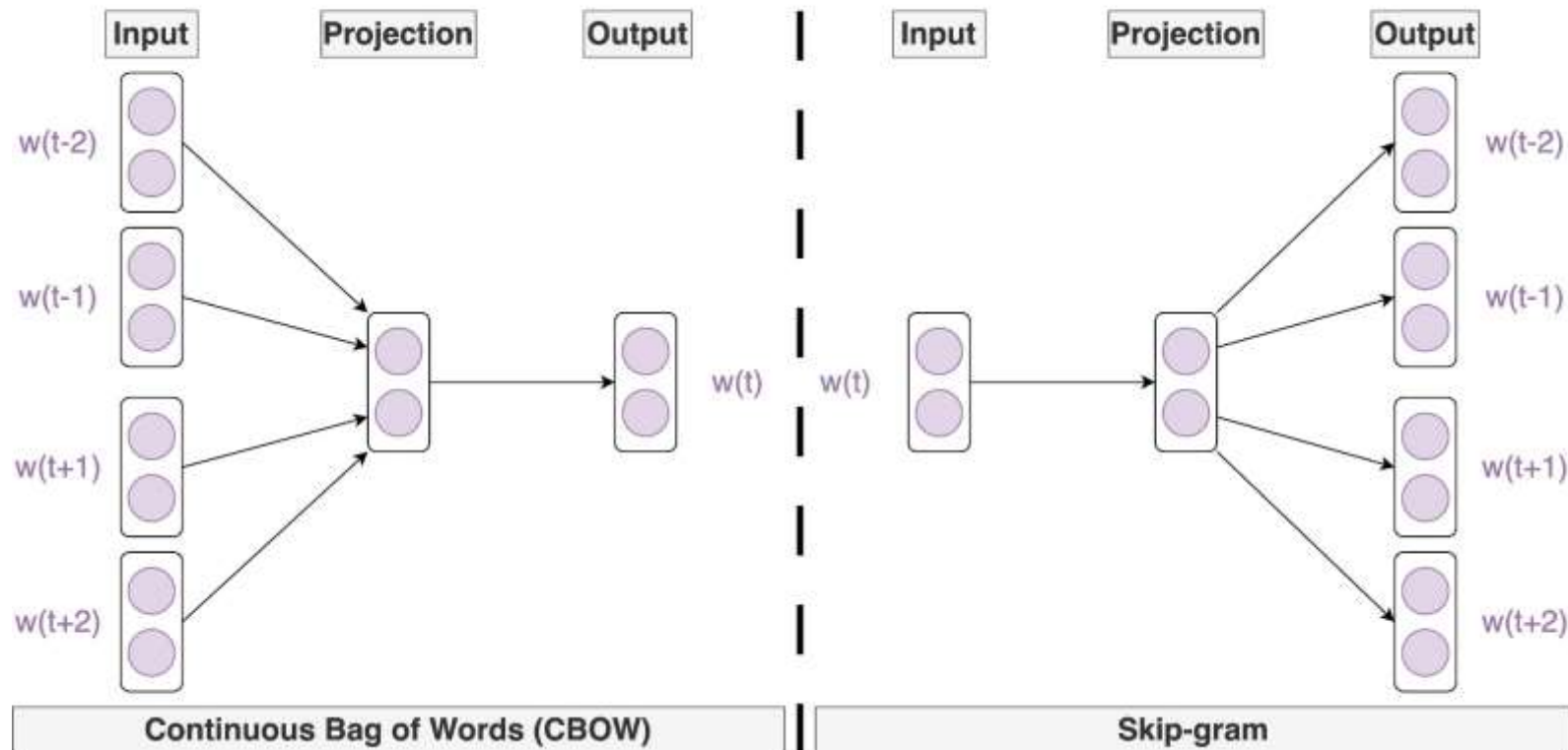
Worst

Sentiment-Specific Word Embedding



Sentiment-Specific Word Embedding

Word2Vec



Sentence Level Sentiment Classification

What is Sentence-Level Classification?

It determines sentiment of a single sentence.

Example

Sentence:

“The camera quality is excellent.”

Prediction:

→ Positive

Methods Used

- CNN
- RNN
- LSTM
- Recursive Neural Networks

Challenges

- Negation handling
- Context understanding
- Sarcasm detection
- Multiple opinions

Convolutional Neural Network (CNN)

What is CNN?

CNN is a deep learning model mainly used for:

- Image processing
- Text classification
- Sentiment analysis

CNN captures important local features.

Example

Sentence:

“The food was absolutely amazing.”

CNN detects:

- “absolutely amazing”
- as strong positive phrase.

Working of CNN in Text

Step 1: Word Embedding

Sentence converted into vectors.

Step 2: Convolution Layer

Detects important patterns.

Step 3: Pooling Layer

Selects strongest features.

Step 4: Fully Connected Layer

Predicts sentiment.

Recurrent Neural Network (RNN) for Sentiment Analysis

Recurrent Neural Network (RNN)

Recurrent Neural Network (RNN) is one of the most important deep learning architectures used for processing sequential data such as text, speech, and time-series information. Unlike traditional neural networks, RNNs have memory capabilities that allow them to remember previous inputs while processing new information. This makes them highly suitable for Natural Language Processing (NLP) tasks such as sentiment analysis.

For example:

Sentence 1:

“I like this movie.”

Sentence 2:

“I do not like this movie.”

Both sentences contain similar words, but the sentiment is completely different because of the word “not.”

Traditional neural networks process words independently and may fail to capture such dependencies. RNN solves this problem by maintaining contextual memory.

Recurrent Neural Network (RNN) for Sentiment Analysis

Working Principle of RNN

RNN processes text sequentially, one word at a time.

Step 1: Input Sentence

The sentence is converted into word embeddings.

Example:

“The product is very good.”

Words are transformed into vectors.

Step 2: Sequential Processing

The network reads words one by one:

- The
- product
- is
- very
- good

At every step, RNN updates its hidden state.

Recurrent Neural Network (RNN) for Sentiment Analysis

Step 3: Hidden State Memory

The hidden state stores information about previous words.

This helps the model understand:

- Context
- Sequence
- Meaning

Step 4: Sentiment Prediction

Finally, the network predicts:

- Positive
- Negative
- Neutral

Mathematical Representation

The hidden state in RNN is calculated as:

$$h_t = f(Wx_t + Uh_{t-1} + b)$$

Where:

- x_t =current input
- h_{t-1} =previous hidden state
- W, U = weight matrices
- b = bias
- f = activation function

Recurrent Neural Network (RNN) for Sentiment Analysis

Advantages of RNN

1. Sequence Understanding

RNN understands sequential information effectively.

2. Context Preservation

Previous words influence future predictions.

3. Better Language Understanding

Useful for:

- Sentiment analysis
- Speech recognition
- Language translation

Recurrent Neural Network (RNN) for Sentiment Analysis

Limitations of RNN

1. Vanishing Gradient Problem

During training, gradients become very small, making learning difficult.

2. Difficulty with Long Sentences

RNN struggles to remember long-term dependencies.

3. Slow Training

Sequential processing increases computation time.

Real-Life Example

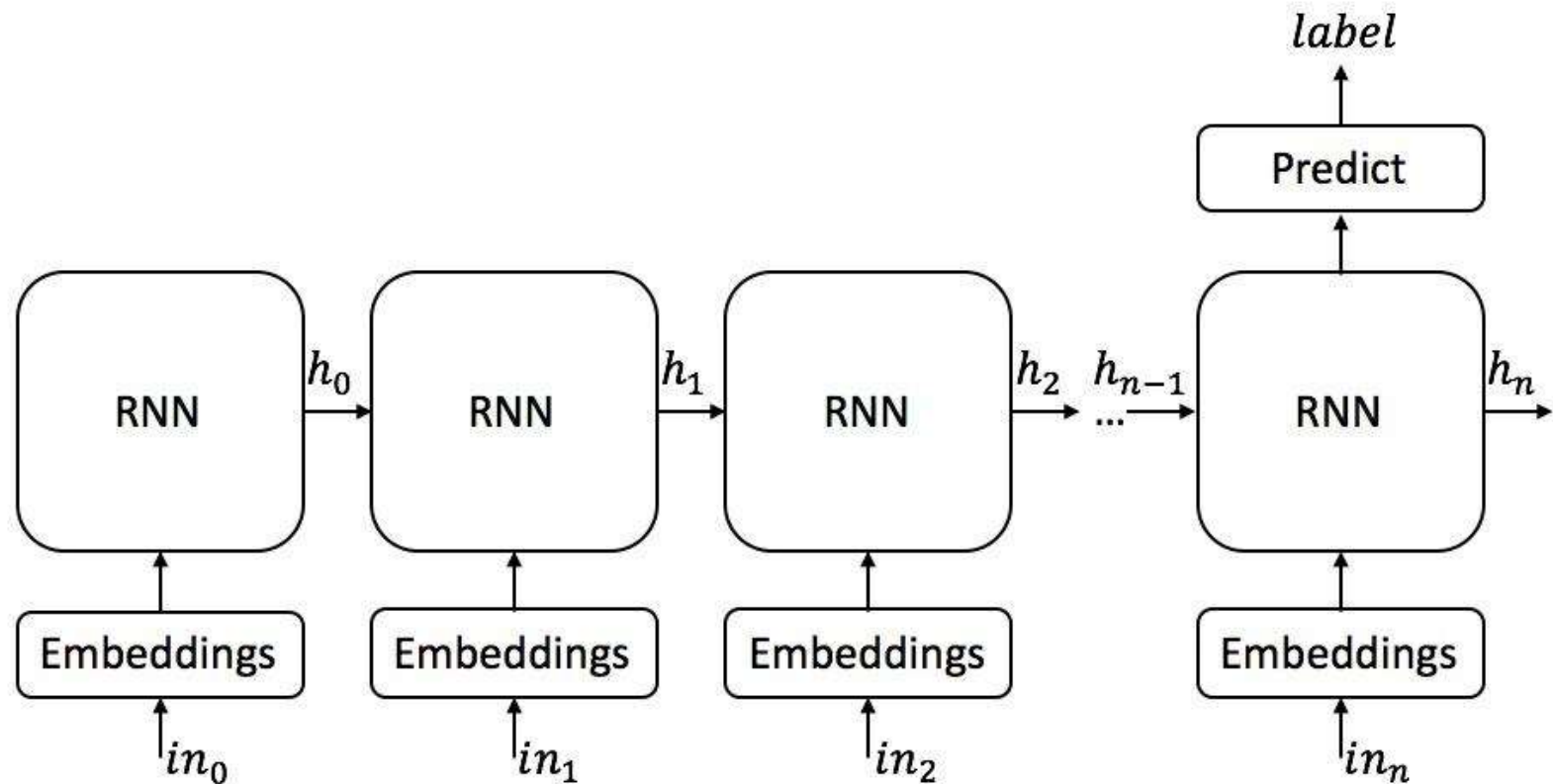
Sentence:

“Although the camera quality is excellent, the battery performance is disappointing.”

RNN can process the sequence and understand:

- Camera → Positive
- Battery → Negative

Recurrent Neural Network (RNN) for Sentiment Analysis



Recursive Neural Network (RecNN)

Recursive Neural Network (RecNN)

Recursive Neural Network is a deep learning architecture designed to process hierarchical and tree-structured data. Unlike RNN, which processes data sequentially, Recursive Neural Networks analyze grammatical structures and syntactic relationships within sentences.

These networks are particularly useful in sentiment analysis because human language often contains nested structures and complex dependencies.

Recursive Neural Network (RecNN)

Working Principle

Recursive Neural Networks use parse trees to represent sentence structure.

Example Sentence:

“The movie was not very good.”

The sentence is broken into smaller components:

- not
- very good
- not very good

The model recursively combines meanings from smaller phrases to larger phrases.

Recursive Neural Network (RecNN)

Parse Tree Structure

The sentence is represented as a hierarchical tree:

- Words form phrases
- Phrases form clauses
- Clauses form complete sentences

This helps the model understand:

- Grammar
- Negation
- Sentence composition

Example Analysis

Sentence:

“The food was not bad.”

Normal interpretation:

- “bad” → negative

Recursive Neural Network interpretation:

- “not bad” → positive

This improves sentiment prediction accuracy.

Recursive Neural Network (RecNN)

Mathematical Representation

Node representation:

$$p = f(W[c_1; c_2] + b)$$

Where:

- c_1, c_2 = child nodes
- W = weight matrix
- b = bias
- f = activation function

Recursive Neural Network (RecNN)

Advantages

1. Better Grammar Understanding

Captures sentence structure effectively.

2. Handles Negation Properly

Useful for complex sentiment analysis.

3. Contextual Interpretation

Understands phrase composition.

Limitations

High Computational Cost

Training parse trees is expensive.

Complex Architecture

More difficult to implement compared to CNN and RNN.

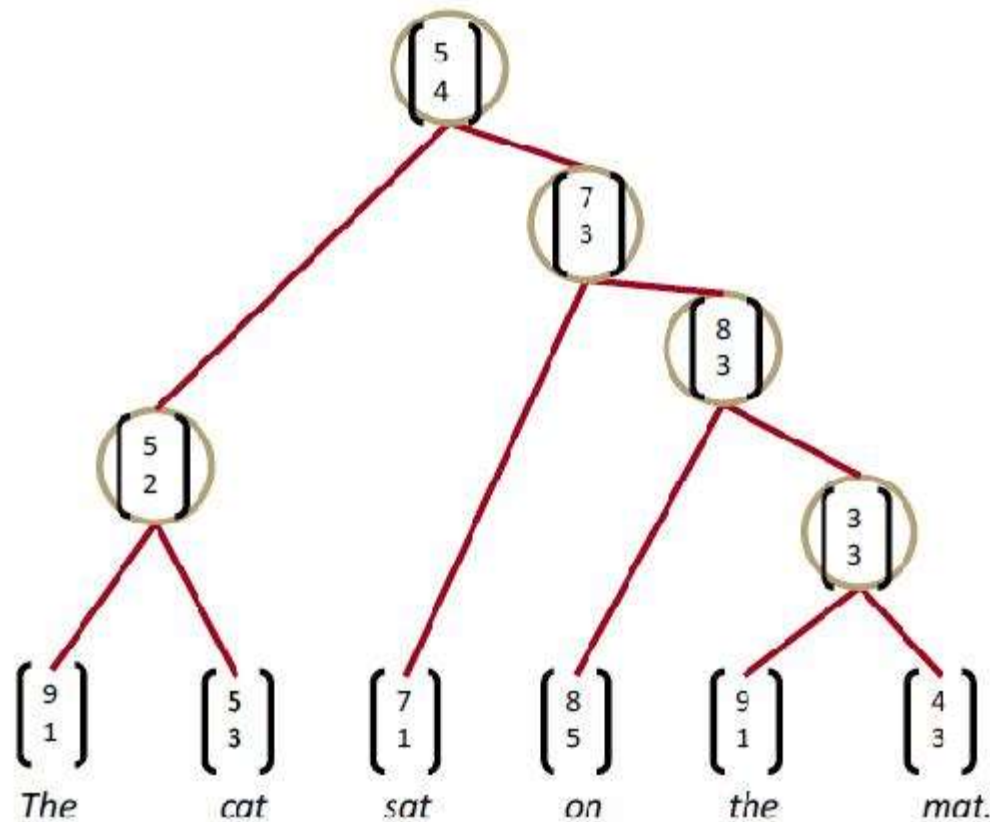
Dependency on Parsing Accuracy

Incorrect parse trees reduce performance.

Applications

- Sentiment analysis
- Syntax analysis
- Semantic understanding
- Question answering systems

Recursive Neural Network (RecNN)



Integration of External Resources

Integration of External Resources in Sentiment Analysis

Deep learning models become more powerful when combined with external knowledge sources. External resources provide additional semantic, emotional, and contextual information that helps improve sentiment classification accuracy.

Deep learning models trained only on datasets may fail to understand domain-specific meanings, rare words, sarcasm, or cultural expressions. External resources help solve these problems.

Integration of External Resources

Types of External Resources

1. Sentiment Lexicons

A sentiment lexicon is a dictionary containing words with associated sentiment scores.

Word	Sentiment
Excellent	Positive
Horrible	Negative

Popular Lexicons:

- SentiWordNet
- VADER
- AFINN

Integration of External Resources

2. Knowledge Graphs

Knowledge graphs store relationships between entities.

Example:

- Apple → Company
- Apple → Fruit

This helps resolve ambiguity.

3. Pretrained Language Models

Models trained on huge datasets:

- BERT
- GPT
- RoBERTa

These models provide contextual embeddings.

4. Ontologies

Ontologies define structured domain knowledge.

Example:

Healthcare ontology understands:

- Fever
- Cold
- Infection

Integration of External Resources

Benefits of External Resources

Better Semantic Understanding

Improves contextual interpretation.

Improved Accuracy

Enhances sentiment prediction performance.

Domain Adaptation

Useful for specialized domains like healthcare and finance.

Sarcasm Handling

Provides contextual clues.

Example

Sentence:

“This virus is spreading rapidly.”

In healthcare:

- “virus” → medical meaning

In cybersecurity:

- “virus” → malware

External resources help identify correct context.

Document-Level Sentiment Classification

Document-Level Sentiment Classification

Document-level sentiment classification analyzes the overall sentiment of an entire document instead of individual sentences. A document may contain multiple opinions, emotions, and viewpoints, making sentiment prediction more challenging.

Examples of documents:

- Product reviews
- News articles
- Research papers
- Blogs
- Social media posts

Document-Level Sentiment Classification

Objective

The goal is to determine whether the complete document expresses:

- Positive sentiment
- Negative sentiment
- Neutral sentiment

Example

Review:

“The camera quality is excellent and the display is beautiful. However, the battery drains quickly.”

The document contains:

- Positive opinions
- Negative opinions

The model predicts the dominant sentiment.

Document-Level Sentiment Classification

Challenges

1. Long Text Processing

Documents may contain thousands of words.

2. Mixed Emotions

Different sections may express different sentiments.

3. Context Preservation

Important information may appear far apart.

Deep Learning Approaches

RNN and LSTM

Capture sequential dependencies.

Attention Mechanism

Focuses on important words.

Transformers

Understand global contextual relationships.

Document-Level Sentiment Classification

Attention Mechanism

Attention helps the model focus on important words.

Example:

“excellent”

“terrible”

These words receive higher importance.

Applications

- Customer feedback analysis
- Political opinion monitoring
- Hotel review analysis
- Social media sentiment tracking

Advantages

- Better understanding of large documents
- Improved contextual analysis
- More accurate predictions

Fine-Grained Sentiment Analysis

Fine-Grained Sentiment Analysis

Fine-grained sentiment analysis provides detailed emotional classification instead of simple positive or negative labels. It measures the intensity and degree of emotions expressed in text.

Traditional sentiment analysis only predicts:

- Positive
- Negative
- Neutral

Fine-grained analysis predicts:

- Very Positive
- Positive
- Neutral
- Negative
- Very Negative

Fine-Grained Sentiment Analysis

Importance

Many real-world applications require detailed emotional understanding.

Example:

- A 5-star review and 4-star review are both positive, but emotional intensity differs.

Example

Sentence:

“The service was absolutely outstanding.”

Prediction:

→ Very Positive

Sentence:

“The service was acceptable.”

Prediction:

→ Neutral

Rating	Sentiment
5 Stars	Very Positive
4 Stars	Positive
3 Stars	Neutral
2 Stars	Negative
1 Star	Very Negative

Fine-Grained Sentiment Analysis

Deep Learning Models Used

- CNN
- RNN
- LSTM
- Transformers

These models analyze:

- Emotion intensity
- Word importance
- Contextual relationships

Advantages

- Detailed emotional understanding
- Better customer insights
- Improved recommendation systems

Applications

E-Commerce

Analyzing customer ratings.

Healthcare

Detecting patient satisfaction.

Education

Analyzing student feedback.

Entertainment

Movie and music review analysis

Challenges

- Subjective interpretation
- Ambiguous language
- Cultural differences

Opinion Mining

Opinion Mining

Opinion Mining is the process of extracting opinions, emotions, attitudes, evaluations, and judgments from textual data. It is closely related to sentiment analysis but focuses more deeply on identifying:

- Who expressed the opinion
- What the opinion is about
- Whether the opinion is positive or negative

Opinion mining helps organizations understand public perception regarding products, services, brands, events, and political issues.

Opinion Mining

Components of Opinion Mining

1. Opinion Holder

The person or entity expressing the opinion.

Example:

“John likes the new iPhone.”

Opinion holder:

→ John

2. Opinion Object

The target being discussed.

Example:

“John likes the new iPhone.”

Opinion object:

→ iPhone

3. Opinion Orientation

The polarity of the opinion.

Example:

→ Positive

Opinion Mining

Example Analysis

Sentence:

“The laptop design is excellent, but the battery backup is poor.”

Feature	Sentiment
Design	Positive
Battery	Negative

Opinion Mining

Types of Opinion Mining

Document-Level

Entire document sentiment.

Sentence-Level

Single sentence sentiment.

Aspect-Level

Feature-specific analysis.

Applications

Business Intelligence

Understanding customer needs.

Political Analysis

Monitoring public opinion.

Healthcare

Patient feedback analysis.

Techniques Used

Machine Learning

- SVM
- Naive Bayes

Deep Learning

- CNN
- RNN
- LSTM
- Transformers

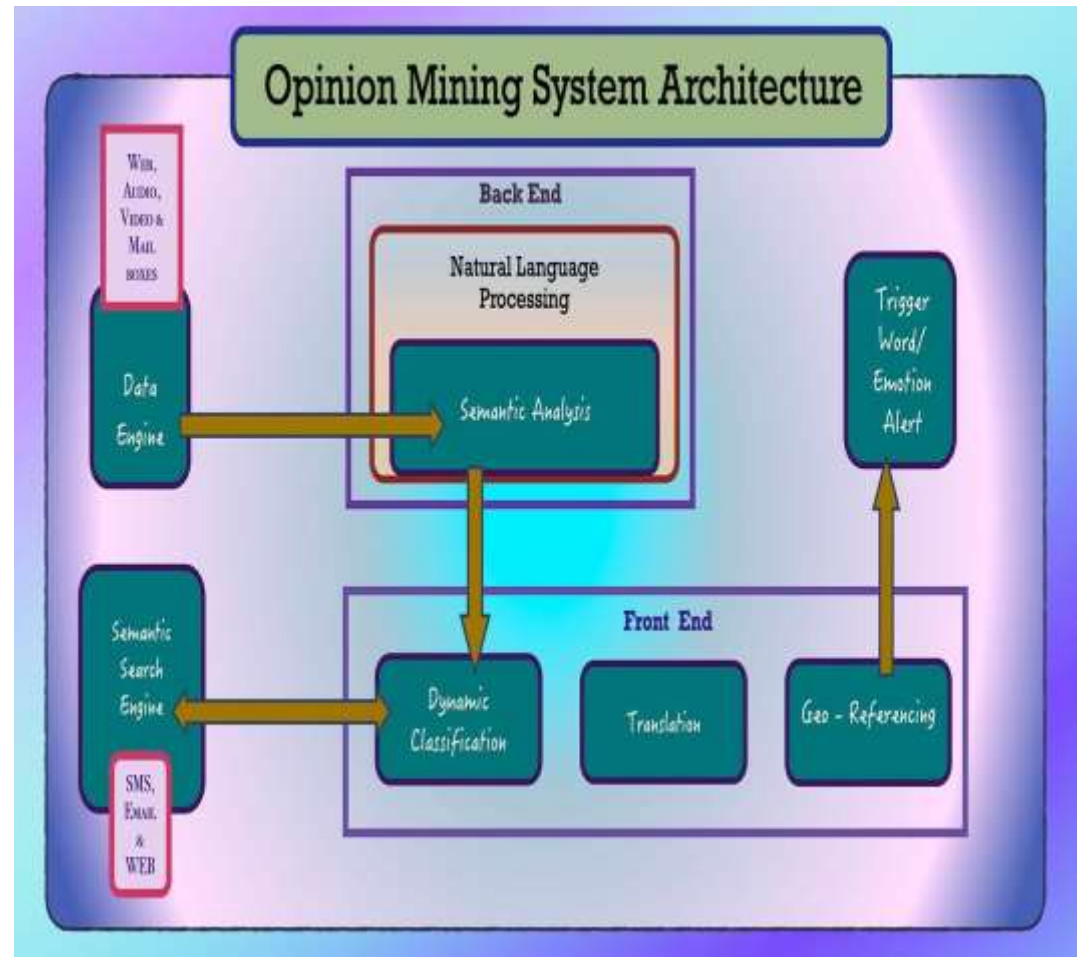
Opinion Mining

Advantages

- Better decision-making
- Improved customer satisfaction
- Market trend analysis
- Brand reputation management

Challenges

- Sarcasm detection
- Fake reviews
- Multilingual data
- Ambiguous expressions



Targeted Sentiment Analysis

Targeted Sentiment Analysis

Targeted Sentiment Analysis is an advanced form of sentiment analysis that identifies the sentiment expressed toward a specific target, entity, or object mentioned in a sentence. Unlike traditional sentiment analysis, which predicts the overall sentiment of the entire sentence or document, targeted sentiment analysis focuses on understanding opinions related to particular targets.

In real-world situations, a single sentence may contain multiple entities with different sentiments. Therefore, identifying the exact sentiment toward each target becomes extremely important.

Targeted Sentiment Analysis

Example

Sentence:

“Samsung phones are excellent, but Apple batteries are disappointing.”

Overall sentiment:

→ Mixed

Target-specific sentiments:

- Samsung phones → Positive
- Apple batteries → Negative

Traditional sentiment analysis may fail to distinguish between these targets, whereas targeted sentiment analysis identifies separate sentiments for each entity.

Targeted Sentiment Analysis

Need for Targeted Sentiment Analysis

In modern digital platforms, users often compare multiple products, services, or individuals within a single review or social media post. Businesses require precise analysis to understand customer opinions regarding specific products or features.

Why It is Important?

- Improves business intelligence
- Helps in competitor analysis
- Enhances recommendation systems
- Provides detailed customer insights

Targeted Sentiment Analysis

Working Process

Step 1: Target Identification

The system identifies important entities or objects mentioned in the text.

Example:

- Samsung
- Apple

Step 2: Context Extraction

The model extracts words surrounding the target.

Example:

- “excellent”
- “disappointing”

Step 3: Sentiment Classification

Deep learning models determine whether the sentiment toward each target is:

- Positive
- Negative
- Neutral

Targeted Sentiment Analysis

Deep Learning Techniques Used

1. Recurrent Neural Networks (RNN)

Understand sequential context.

2. Long Short-Term Memory (LSTM)

Capture long-range dependencies.

3. Attention Mechanism

Focus on important target-related words.

4. Transformer Models

Models like BERT provide contextual understanding.

Example Analysis

Sentence:

“The camera quality of the phone is amazing, but the battery life is terrible.”

Target-wise output:

Target	Sentiment
Camera Quality	Positive
Battery Life	Negative

Targeted Sentiment Analysis

Applications

E-Commerce

Analyzing customer opinions about product features.

Social Media Monitoring

Tracking public opinion toward brands or celebrities.

Political Analysis

Understanding opinions about politicians or policies.

Customer Support Systems

Identifying customer complaints about specific services.

Targeted Sentiment Analysis

Advantages

- Provides detailed sentiment insights
- Better business decision-making
- Accurate customer feedback analysis
- Improves marketing strategies

Challenges

1. Ambiguity

Same sentence may contain multiple targets.

2. Context Dependency

Sentiment depends on surrounding words.

3. Sarcasm Handling

Difficult to identify actual emotions.

4. Complex Grammar

Long sentences create processing challenges.

Real-Life Example

Amazon uses targeted sentiment analysis to determine:

- Which product features customers like
- Which features customers dislike

This helps companies improve product quality.

Aspect Level Sentiment Analysis (ABSA)

Aspect Level Sentiment Analysis

Aspect Level Sentiment Analysis (ABSA), also known as Feature-Based Sentiment Analysis, is a fine-grained sentiment analysis technique that identifies sentiments related to specific aspects or features of a product, service, or topic. Instead of determining only the overall sentiment, ABSA focuses on understanding opinions about individual components or attributes. For example, a customer may like the design of a mobile phone but dislike its battery performance. Aspect-level analysis captures these detailed opinions separately.

Example

Sentence:

“The laptop design is beautiful, but the performance is very slow.”

Aspect Level Sentiment Analysis (ABSA)

Aspect-wise sentiments:

Aspect	Sentiment
Design	Positive
Performance	Negative

Importance of ABSA

Modern businesses need detailed customer feedback rather than overall ratings.

ABSA helps companies identify:

- Strengths of products
- Weaknesses of products
- Customer expectations
- Areas requiring improvement

Aspect Level Sentiment Analysis (ABSA)

Steps in Aspect Level Sentiment Analysis

Step 1: Aspect Extraction

Identify important aspects/features.

Example:

- Design
- Performance

Step 2: Sentiment Detection

Determine sentiment associated with each aspect.

Step 3: Classification

Classify sentiment as:

- Positive
- Negative
- Neutral

Aspect Level Sentiment Analysis (ABSA)

Types of Aspects

Explicit Aspects

Directly mentioned in the sentence.

Example:

“Battery life is excellent.”

Aspect:

→ Battery life

Implicit Aspects

Not directly mentioned but implied.

Example:

“It heats up quickly.”

Implied aspect:

→ Processor/System performance

Aspect Level Sentiment Analysis (ABSA)

Deep Learning Models Used

CNN

Captures local features.

RNN/LSTM

Understands sequential relationships.

Attention-Based Networks

Focus on aspect-related words.

BERT and Transformers

Provide contextual semantic understanding.

Aspect Level Sentiment Analysis (ABSA)

Applications

Product Review Analysis

Amazon and Flipkart analyze feature-wise customer opinions.

Hotel Reviews

Analyzing:

- Food
- Service
- Cleanliness
- Staff behavior

Healthcare

Analyzing patient feedback regarding hospital services.

Restaurant Reviews

Understanding customer satisfaction for:

- Taste
- Ambience
- Service

Aspect Level Sentiment Analysis (ABSA)

Advantages

- Detailed sentiment insights
- Better product improvement strategies
- Enhanced customer satisfaction
- Accurate recommendation systems

Challenges

1. Aspect Extraction Difficulty

Implicit aspects are hard to identify.

2. Multiple Aspects in One Sentence

Complex sentence structures create challenges.

3. Sarcasm and Irony

Difficult for models to interpret.

4. Domain Dependency

Different domains use different vocabulary.

Aspect Level Sentiment Analysis (ABSA)

Real-Life Example

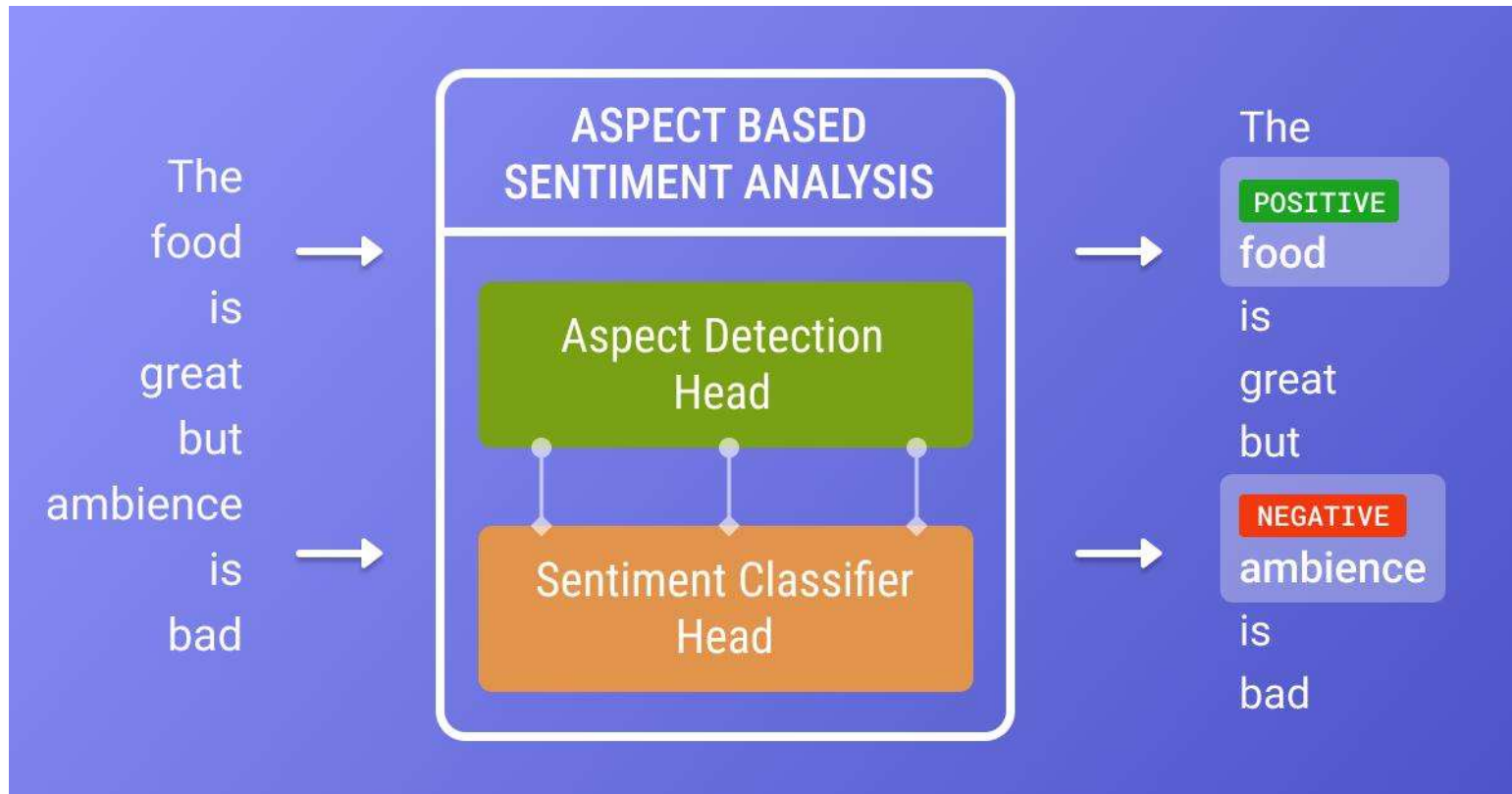
Restaurant Review:

“The food was delicious, but the service was extremely slow.”

Aspect-wise results:

Aspect	Sentiment
Food	Positive
Service	Negative

Aspect Level Sentiment Analysis (ABSA)



Stance Detection

Stance Detection

Stance Detection is a Natural Language Processing (NLP) task that determines whether a text expresses:

- Support
- Opposition
- Neutrality

toward a specific topic, claim, event, or target.

Unlike sentiment analysis, stance detection focuses on identifying the writer's position regarding a subject rather than just detecting emotions.

A sentence may contain positive language but still oppose a topic, making stance detection more complex than standard sentiment analysis.

Stance Detection

Example

Topic:

“Online Education”

Sentence:

“Online classes save travel time and improve accessibility.”

Stance:

→ Favor

Another sentence:

“Online education reduces practical learning experiences.”

Stance:

→ Against

Stance Detection

Difference Between Sentiment Analysis and Stance Detection

Sentiment Analysis

Detects emotion

Positive/Negative/Neutral

Focus on feelings

Stance Detection

Detects viewpoint

Favor/Against/Neutral

Focus on opinion toward topic

Importance of Stance Detection

Stance detection is extremely useful for:

- Political analysis
- Fake news detection
- Social media monitoring
- Public opinion tracking

Stance Detection

Working Process

Step 1: Topic Identification

Identify the target topic.

Step 2: Context Understanding

Analyze relationship between text and topic.

Step 3: Stance Classification

Classify as:

- Favor
- Against
- Neutral

Deep Learning Models Used

RNN and LSTM

Understand sequential context.

Attention Mechanism

Focus on important stance-related words.

Transformer Models (BERT)

Capture semantic relationships.

Stance Detection

Applications

Political Analysis

Understanding public support for political leaders.

Social Media Monitoring

Tracking opinions on social issues.

Fake News Detection

Identifying misleading narratives.

Brand Monitoring

Understanding customer stance toward brands.

Challenges

1. Implicit Opinions

Stance may not be directly expressed.

2. Sarcasm

Irony changes actual meaning.

3. Ambiguous Language

Different interpretations possible.

4. Context Dependency

Understanding requires background knowledge.

Stance Detection

Real-Life Example

Topic:

“Artificial Intelligence in Education”

Sentence:

“AI can personalize learning and improve student performance.”

Stance:

→ Favor

Stance Detection

#	Tweet	Target	Sentiment	Stance
1	It is so much fun having younger friends who are expecting babies. #beentheredonethat #chooselife .	Legalisation of Abortion	+	-
2	Life is sacred on all levels. Abortion does not compute with my philosophy. (Red on #OITNB)	Legalization of Abortion	0	-
3	The biggest terror threat in the World is climate change #drought #floods	Climate Change is the real concern	-	+
4	I am sad that Hillary lost this presidential race	Hillary Clinton	-	+

Sarcasm Recognition

Sarcasm Recognition

Sarcasm Recognition is one of the most challenging tasks in sentiment analysis and Natural Language Processing. Sarcasm occurs when a person says something positive but actually means something negative, or vice versa.

Humans easily understand sarcasm using tone, facial expressions, and context, but computers struggle because sarcastic sentences often contain contradictory meanings.

Example

Sentence:

“Wonderful! My internet stopped working during the meeting.”

Literal meaning:

→ Positive

Actual sentiment:

→ Negative

Traditional sentiment analysis systems may incorrectly classify such sentences because they focus only on positive words like “Wonderful.”

Sarcasm Recognition

Importance of Sarcasm Recognition

Without sarcasm detection:

- Sentiment analysis becomes inaccurate
- Customer feedback interpretation becomes misleading
- Social media monitoring produces incorrect results

Sarcasm recognition improves the reliability of AI systems.

Characteristics of Sarcasm

1. Opposite Meaning

Actual meaning differs from literal meaning.

2. Emotional Contrast

Positive words express negative emotions.

3. Context Dependence

Requires understanding of situation and background.

Sarcasm Recognition

Types of Sarcasm

Self-Deprecating Sarcasm

“I am such a genius; I forgot my own password again.”

Polite Sarcasm

“Thanks for replying after three days.”

Angry Sarcasm

“Fantastic service! The order arrived broken.”

Challenges in Sarcasm Recognition

1. Lack of Tone Information

Text lacks voice expressions.

2. Context Requirement

Understanding often depends on previous conversation.

3. Cultural Differences

Sarcasm varies across cultures and languages.

4. Ambiguous Language

Literal words conflict with actual intent.

Sarcasm Recognition

Deep Learning Techniques Used

CNN

Detects important local phrases.

RNN/LSTM

Understands contextual relationships.

Attention Mechanism

Focuses on contradictory words.

Transformer Models

BERT and GPT improve contextual understanding.

Example Analysis

Sentence:

“Great! Another assignment deadline.”

Detected clues:

- “Great” appears positive
- Context indicates frustration

Final sentiment:

→ Negative/Sarcastic

Sarcasm Recognition

Applications

Social Media Monitoring

Understanding real user emotions.

Chatbots

Improving conversational intelligence.

Customer Feedback Systems

Avoiding incorrect sentiment interpretation.

Mental Health Analysis

Detecting emotional distress hidden behind sarcasm.

Advantages of Sarcasm Recognition

- Improves sentiment accuracy
- Better emotion understanding
- More intelligent AI systems
- Enhanced customer experience analysis

Real-Life Example

Twitter users frequently use sarcasm:

“Perfect weather for getting stuck in traffic.”

Actual sentiment:

→ Negative

Sarcasm recognition helps AI interpret such expressions correctly.

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